

Software Engineering – Model Solutions

Scenario 1: Hospital Management System

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - `authenticateUser(user_id, credentials)` - `initiateCoreProcess(entity_id, parameters)` - `updateSystemState(entity_id, state_data)` - `terminateProcess(entity_id, status)` - `generateReport(entity_id, metrics)` Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 2: Autonomous Car Driving Simulator

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - authenticateUser(user_id, credentials) - initiateCoreProcess(entity_id, parameters) - updateSystemState(entity_id, state_data) - terminateProcess(entity_id, status) - generateReport(entity_id, metrics) Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 3: Online Banking & Fraud Detection System

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - authenticateUser(user_id, credentials) - initiateCoreProcess(entity_id, parameters) - updateSystemState(entity_id, state_data) - terminateProcess(entity_id, status) - generateReport(entity_id, metrics) Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 4: Railway Reservation System

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - authenticateUser(user_id, credentials) - initiateCoreProcess(entity_id, parameters) - updateSystemState(entity_id, state_data) - terminateProcess(entity_id, status) - generateReport(entity_id, metrics) Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 5: Smart City Traffic Control System

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - authenticateUser(user_id, credentials) - initiateCoreProcess(entity_id, parameters) - updateSystemState(entity_id, state_data) - terminateProcess(entity_id, status) - generateReport(entity_id, metrics) Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 6: Military Drone Command Platform

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - authenticateUser(user_id, credentials) - initiateCoreProcess(entity_id, parameters) - updateSystemState(entity_id, state_data) - terminateProcess(entity_id, status) - generateReport(entity_id, metrics) Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 7: University LMS

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - authenticateUser(user_id, credentials) - initiateCoreProcess(entity_id, parameters) - updateSystemState(entity_id, state_data) - terminateProcess(entity_id, status) - generateReport(entity_id, metrics) Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 8: Air Traffic Control System

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - authenticateUser(user_id, credentials) - initiateCoreProcess(entity_id, parameters) - updateSystemState(entity_id, state_data) - terminateProcess(entity_id, status) - generateReport(entity_id, metrics) Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 9: Disaster Response Coordination System

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - `authenticateUser(user_id, credentials)` - `initiateCoreProcess(entity_id, parameters)` - `updateSystemState(entity_id, state_data)` - `terminateProcess(entity_id, status)` - `generateReport(entity_id, metrics)` Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.

Scenario 10: E-commerce Platform

Q1(a) Lifecycle Model: Spiral or Incremental-Iterative model is suitable due to evolving requirements, risk handling, and modular deployment. Q1(b) Major Functions: - `authenticateUser(user_id, credentials)` - `initiateCoreProcess(entity_id, parameters)` - `updateSystemState(entity_id, state_data)` - `terminateProcess(entity_id, status)` - `generateReport(entity_id, metrics)` Q2(a) Additional Modules: - Logging & Monitoring Module - Security & Access Control - Data Analytics Engine - Notification System - Backup & Recovery Module Q2(b) Sequence Flow (Core Use Case): User → Authentication → Core Service → Database → Response → Report Generation Q3 Risks: Technical: System scalability, integration complexity. Operational: User resistance, training gaps. Project: Scope creep, budget overrun. Mitigation: Prototyping, phased rollout, risk reviews. Q4(a) Test Cases: 1. Valid input → Successful execution. 2. Invalid credentials → Access denied. 3. Boundary data values → Correct validation. 4. Network failure → Graceful recovery. 5. High load → Performance maintained. Q4(b) COCOMO Estimation (Example 220 KLOC, Semi-detached): $\text{Effort} = 3.0 \times (220)^{1.12} \approx 950$ Person-Months $\text{TDEV} = 2.5 \times (950)^{0.35} \approx 30$ Months Justification: Moderate complexity, mixed experience team.